Ryan Rubenzahl

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Education

Ph.D., California Institute of Technology **Pasadena, CA**
Astrophysics, Advisor: Andrew Howard. Defended May 13, 2024. *2018–2024*
Thesis: *From the Sun to the Stars: A Solar Calibrator for the Keck Planet Finder And New Frontiers in Exoplanet Obliquities*.

M.S., California Institute of Technology **Pasadena, CA**
Astrophysics. *2021*

B.S., University of Rochester **Rochester, NY**
Physics and Astronomy. Magna cum laude with highest distinction. Minor: Math. *2014–2018*
Senior thesis: *Identifying Type Ia Supernovae in Extragalactic Spectra*. Advisor: Segev BenZvi.

Academic Employment

Flatiron Institute **Center for Computational Astrophysics**
Flatiron Research Fellow *2024–*
Developing statistical and computational tools for comprehensive multi-instrument cross-comparisons of EPRV Sun-as-a-star datasets to characterize stellar variability and instrumental drift towards 10 cm s^{-1} precision.

California Institute of Technology **Department of Astronomy**
Graduate Researcher *2018–2024*
Built an autonomous solar feed for the Keck Planet Finder; characterized exoplanet masses, orbits, and obliquities; helped select targets, organize observations, and model timeseries for the TESS-Keck Survey.

University of Rochester **Physics & Astronomy Department**
Undergraduate Researcher *2014–2018*
Simulated spectra of type-Ia supernova host galaxies to feed a random forest classifier to estimate DESI yield; searched for and analyzed TeV phenomena with HAWC, such as "Fermi Bubbles" in M31 and emission from compact binaries

Appointments and Experience

American Museum of Natural History **Department of Astrophysics**
Research Associate, Division of Physical Sciences *2026–*
Advising summer undergradutes and CUNY masters students on projects related to PARVI solar data and flares.

W. M. Keck Observatory **Keck Visiting Scholars Program**
Keck Visiting Scholar *2022*
Developed automated state machine control software and installed the KPF Solar Calibrator at Keck Observatory.

Awards and Honors

2024: Three Minute Thesis Competition, *People's Choice Award*, Caltech
2023: Rodger Doxsey Travel Prize, American Astronomical Society
2018: NSF Graduate Research Fellowship, National Science Foundation
2018: Stoddard Prize, *best senior thesis in Physics & Astronomy*, U. of Rochester
2018: Janet Fogg Prize, *awarded to one senior for dedicated departmental service*, U. of Rochester
2018: Undergraduate Teaching Award, U. of Rochester
2017: Barry M. Goldwater Scholarship, Goldwater Foundation
2017: Award for Excellence in Programming, *for "Earth Hour"*, U. of Rochester
2016, 2017: Continuing Student Scholarship, U. of Rochester
2014: Bausch and Lomb Honorary Science Award & Scholarship, U. of Rochester

Awarded Telescope Time

as PI

Illuminating the Origins of Hot Jupiters by Directly Measuring Tidal Orbital Evolution **1.5 nights**
PI: NASA/Keck program N127 (0.75 nights), Co-I: Caltech/Keck program C333 (0.75 nights) *$10,750*
2026A: KPF, Keck-I (10-m)

Probing Hot Jupiter and Polar Neptune Formation with KPF **2.25 nights**
PI†: Caltech/Keck program C399
2023B: KPF, Keck-I (10-m)

KPF Science Verification: New Frontiers in Exoplanet Obliquities **3.5 nights**
PI†: Caltech/Keck program C013
2023A: KPF, Keck-I (10-m)
†Functionally PI, but officially "PI: Howard" as Caltech grad students cannot PI Keck proposals.

as Co-I

Measuring the Mass of the Youngest Transiting Exoplanet **33.8 hr**
Co-I: GN-2026B-Q-224 (PI: Tran)
2026B: MAROON-X, Gemini N (8-m)

KPF Science Verification: Transit Spectroscopy of Ultra-Hot Jupiters **2 nights**
Co-I: Caltech/Keck program C400 (PI: Howard)
2023B: KPF, Keck-I (10-m)

Major Grants and Competitive Funding Awards

Keck Planet Finder Stellar Obliquity Survey **$663,394 (PI: Howard)**
NASA Exoplanets Research Program *2024-2026*

Student Mentorship

Lyniya Edwards
Masters Student, The City University of New York *2025-*
For her Master's Thesis, Lyniya is investigating solar flares using EPRV solar data (KPF, NEID) to construct a flare-sensitive line list and probe flare propagation throughout the solar amtmosphere.

Kate Hobbs
Ph.D. Student, Queen's University Belfast *2025-*
As part of the CCA Predoc program, Kate combined solar data from KPF, NEID, EXPRES, and HARPS-N to isolate characteristic modes of line shape distortions with PCA, and connected these to surface features resolved by SDO.

Xuezhen Li
Undergraduate, Pasadena City College → Aerospace Engineering major at UC Berkeley *2023*
As part of the Caltech Connection program, Xuezhen researched the exoplanet measurement techniques, learned Python, and explored fitting real radial velocity and transit data to discover an exoplanet.

Jared Siegel
Undergraduate, University of Chicago → current Ph.D. student at Princeton *2021*
As a Caltech SURF student, Jared wrote a line-by-line radial velocity pipeline and derived a novel activity indicator based on spectral line depth, which outperformed traditional activity indicators in reducing the RMS of αCenB HARPS RVs.

Aanica Gonzales-Rogers
Undergraduate, California Institute of Technology → Neurscience Ph.D. student at UC Berkeley *2020*
As a Caltech SURF student, Aanica explored the detectability of exoplanets in wide orbits using `orbitize!` and simulated astrometric measurements anticipated in future Gaia data releases.

Teaching Positions

TA, Astronomy Department, California Institute of Technology

2020: Ay 124: Structure and Evolution of Galaxies, Ay 105: Optical Instrumentation Lab (remote)

2019: Ay 122a: Optical/IR Measurements and Instrumentation

TA, Physics and Astronomy Department, University of Rochester

2018: AST 142: Elementary Astrophysics (Honors)

2017: AST 111: The Solar System & Its Origin, AST 142: Elementary Astrophysics

2016: PHY 141 Lab: Mechanics (Honors), AST 102: Relativity, Black Holes, and the Big Bang

2015: AST 106: Cosmic Origins of Life, AST 104: The Solar System

Leadership Roles and Collaborations

2024-: *The Terra Hunting Survey*, a ten year nightly survey of nearby Sun-like stars with HARPS-3. Science Team member (Lead: Planet Confidence subgroup)

2020-: *The Keck Planet Finder*, an ultra-stabilized EPRV system for W. M. Keck Observatory. Instrument Team member (Lead: Solar Calibrator), Science Team member.

2019-2024: *The* TESS-*Keck Survey*, a survey with Keck/HIRES to measure masses of 100+ TESS candidates. Graduate student member (target selection, timeseries modeling, scheduling).

Refereeing and Proposal Review

Referee: Astronomy & Astrophysics, the Astronomical Journal, the Astrophysical Journal Letters

Invited review panelist: NASA ROSES Astrophysics

Service

Sun-as-a-Star Part I: Accessing and Working with EPRV Solar Datasets — **EPRV 6**

Splinter Session Organizer — *2025*

Organized and created materials for a session containing a suite of tutorials and contributed talks to empower the community to work with the EPRV solar datasets.

Respect is a Part of Research — **California Institute of Technology**

Organizer/facilitator — *2021–2023*

Presented material and facilitated group discussions for a peer-lead sexual harassment/sexual assault prevention workshop for incoming first year graduate students during orientation.

Code/Astro

Teaching Assistant — *2021–2022*

Assisted students with writing their Python packages throughout the software development workshop.

FUTURE — **California Institute of Technology**

Volunteer — *2019–2021*

Co-ran a CV/SoP writing workshop for undergraduate women and nonbinary students

Intro2Astro

Teaching Assistant — *2019–2020*

Presented lessons on Bayesian statistics, model fitting, and model comparison.

College Center for Advising Services — **University of Rochester**

Peer Advisor — *2018*

Advised students with their majors, course selection, research opportunities, networking, independent study, study abroad, applying for grad school, and more.

C.E.K. Mees Observatory — **University of Rochester**

Tour Guide — *2015–2018*

Presented the history of the observatory, current research at UR, and operated the 24-inch telescope.

Selected Presentations

Conferences

2026: GPRV 2 (contributed talk) — Oxford, UK

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2026: Exoplanets 6 (contributed talk) — Porto, Portugal
2026: Cool Stars 23 (contributed splinter-session talk) — Tokyo, Japan
2025: EPRV 6 (contributed talk) — Porto, Portugal
2025: Know Thy Star, Know Thy Planet 2 (poster) — Pasadena, CA
2024: Extreme Solar Systems V (contributed talk) — Christchurch, New Zealand
2024: AAS 243 (dissertation talk) — New Orleans, LA
2023: Keck Science Meeting (contributed talk) — Berkeley, CA
2023: EPRV 5 (poster) — Santa Barbara, CA
2023: Sun-as-a-star workshop (invited talk) — Flatiron Institute
2023: AAS 241 (contributed talk #323.01) — Seattle, WA
2022: ERES VII (contributed talk) — The Pennsylvania State University
2020: AAS 235 (talk) — Honolulu, HI
2019: Keck Science Meeting (contributed talk) — UCLA
2019: Extreme Solar Systems IV (poster #309.08) — Reykjavik, Iceland
2018: AAS 231 (poster #250.05) — National Harbor, MD
2016, 2017: Rochester Symposium for Physics Students (talk) — Rochester, NY

Seminars

2026: Journal Club (invited talk) — University of Pennsylvania
2026: Lunch Seminar (invited talk) — Carnegie Observatories
2026: Special Seminar (invited talk) — JPL
2026: Special Seminar (invited talk) — UCSC
2023: Exoplanets Seminar (invited talk) — Princeton University
2023: Dix Planetary Science Seminar (talk) — Caltech

Press and Public Media

2024: AAS Journal Author Series, interviewed on our measurement of KELT-18 b (**Rubenzahl**+24c, AJ)

2023: "Is there life beyond Earth? The Keck Planet Finder could tell us.", interviewed for article on KPF and the Solar Calibrator for Dell Technologies Perspectives (**Rubenzahl**+23, PASP).

2022: "Keck Observatory's Newest Planet Hunter Puts Its Eye on the Sky", Caltech Press Release.

2021: "The Strange Case of the Misplaced Cotton Candy World", interviewed for article on WASP-107 b for supercluster.com (**Rubenzahl**+21, AJ; Piaulet, Benneke, **Rubenzahl**+21, AJ).

2020: "Measuring the Speed of Stars More Precisely Than Your Car's Speedometer", guest speaker for the San Diego Astronomy Association.

2020: "Measuring the Speed of Stars", presented a public talk for CaltechAstro outreach on YouTube.

Publications

Refereed: 6 first author (118 citations), 5 second/third author (131 citations), 59 nth author (1,936 citations)
†Not refereed: 1 arXiv preprint (15 citations), 1 research note (4 citations)

Selected publications (249 citations)

2026: J. K. Luhn, **R. A. Rubenzahl**, S. Halverson, & L. L. Zhao, "Exposure-averaged Gaussian Processes for Combining Overlapping Datasets", *The Astronomical Journal*, 171, 342 [1 citation]

2026: **R. A. Rubenzahl**, S. Hattori, S. Särkkä, W. M. Farr, J. K. Luhn, M. Bedell, & D. Foreman-Mackey, "Scalable Gaussian Processes for Integrated and Overlapping Measurements via Augmented State-space Models", *The Astronomical Journal*, 171, 259 [2 citations]

2025: S. Giacalone, A. W. Howard, **R. A. Rubenzahl**, F. Dai, L. B. Handley, H. Isaacson, S. Halverson, M. Brodheim, M. Brown, T. W. Carmichael, et al., "A Hot Jupiter with a Retrograde Orbit Around A Sun-like Star and a Toy Model of Hot Jupiters in Wide Binary Star Systems", *Publications of the Astronomical Society of the Pacific*, 137, 074401 [5 citations]

2025: L. B. Handley, A. W. Howard, **R. A. Rubenzahl**, F. Dai, D. Tyler, R. A. Lee, S. Giacalone, H. Isaacson, B. Fulton, A. Householder, et al., "An Obliquity Measurement of the Hot Neptune TOI-1694b", *The Astronomical Journal*, 169, 212 [14 citations]

2024: **R. A. Rubenzahl**, F. Dai, S. Halverson, A. W. Howard, A. Householder, B. Fulton, A. Behmard, S. R. Gibson, A. Roy, A. P. Shaum, et al., "KPF Confirms a Polar Orbit for KELT-18 b", *The Astronomical Journal*, 168, 188 [10 citations]

2024: **R. A. Rubenzahl**, A. W. Howard, S. Halverson, C. Petrovich, I. Angelo, G. Stefánsson, F. Dai, A. Householder, B. Fulton, S. R. Gibson, et al., "Obliquity Constraints for the Extremely Eccentric Sub-Saturn Kepler-1656 b", *The Astrophysical Journal Letters*, 971, L40 [8 citations]

2024: **R. A. Rubenzahl**, F. Dai, A. W. Howard, J. J. Lissauer, J. Van Zandt, C. Beard, S. Giacalone, J. M. A. Murphy, A. Chontos, J. Lubin, et al., "The TESS-Keck Survey. XII. A Dense 1.8 $R_\oplus$ Ultra-short-period Planet Possibly Clinging to a High-mean-molecular-weight Atmosphere after the First Gigayear", *The Astronomical Journal*, 167, 153 [12 citations]

2023: **R. A. Rubenzahl**, S. Halverson, J. Walawender, G. M. Hill, A. W. Howard, M. Brown, E. Ida, J. Tehero, B. J. Fulton, S. R. Gibson, et al., "Staring at the Sun with the Keck Planet Finder: An Autonomous Solar Calibrator for High Signal-to-noise Sun-as-a-star Spectra", *Publications of the Astronomical Society of the Pacific*, 135, 125002 [31 citations]

2022: J. C. Siegel, **R. A. Rubenzahl**, S. Halverson, & A. W. Howard, "Into the Depths: A New Activity Metric for High-precision Radial Velocity Measurements Based on Line Depth Variations", *The Astronomical Journal*, 163, 260 [13 citations]

2021: **R. A. Rubenzahl**, F. Dai, A. W. Howard, A. Chontos, S. Giacalone, J. Lubin, L. J. Rosenthal, H. Isaacson, N. M. Batalha, I. J. M. Crossfield, et al., "The TESS-Keck Survey. IV. A Retrograde, Polar Orbit for the Ultra-low-density, Hot Super-Neptune WASP-107b", *The Astronomical Journal*, 161, 119 [55 citations]

2021: C. Piaulet, B. Benneke, **R. A. Rubenzahl**, A. W. Howard, E. J. Lee, D. Thorngren, R. Angus, M. Peterson, J. E. Schlieder, M. Werner, et al., "WASP-107b's Density Is Even Lower: A Case Study for the Physics of Planetary Gas Envelope Accretion and Orbital Migration", *The Astronomical Journal*, 161, 70 [98 citations]

Publications contributed to as collaborator (192 citations)

2026: K. L. Hobbs, C. A. Watson, J. C. Costes, Y. Unruh, D. C. Yaptangco, et al. **(incl. RAR)**, "The Fe I 4377 Å line as a solar faculae indicator: insights from spectral ratio analysis", *Monthly Notices of the Royal Astronomical Society*, 551, stag1462

2026: S. Blunt, J. Wang, R. Murray-Clay, B. Macintosh, **R. A. Rubenzahl**, et al., "Evidence for a Peak at $\sim$0.3 in the Eccentricity Distribution of Typical Super-Jovian Exoplanets", *The Astrophysical Journal Letters*, 999, L26 [6 citations]

2025: A. Behmard, C. L. Brinkman, S. Hattori, **R. A. Rubenzahl**, & M. Bedell, "A Link between Rocky Planet Density and Host Star Chemistry", *The Astronomical Journal*, 170, 282 [8 citations]

2025: E. Y. Zhang, H.-Y. Teng, F. Dai, A. W. Howard, S. P. Halverson, et al. **(incl. RAR)**, "TOI-880 is an Aligned, Coplanar, Multiplanet System", *The Astronomical Journal*, 170, 175 [5 citations]

2025: A. W. Howard, E. Sinukoff, S. Blunt, E. A. Petigura, I. J. M. Crossfield, et al. **(incl. RAR)**, "Planet Masses, Radii, and Orbits from NASA's K2 Mission", *The Astrophysical Journal Supplement Series*, 278, 52 [24 citations]

2024: M. Hon, D. Huber, Y. Li, T. S. Metcalfe, T. R. Bedding, et al. **(incl. RAR)**, "Asteroseismology of the Nearby K Dwarf σ Draconis Using the Keck Planet Finder and TESS", *The Astrophysical Journal*, 975, 147 [22 citations]

2024: S. Giacalone, F. Dai, J. J. Zanazzi, A. W. Howard, C. D. Dressing, et al. **(incl. RAR)**, "The OATMEAL

Survey. I. Low Stellar Obliquity in the Transiting Brown Dwarf System GPX-1", *The Astronomical Journal*, 168, 189 [13 citations]

2024: H.-Y. Teng, F. Dai, A. W. Howard, H. Isaacson, **R. A. Rubenzahl**, et al., "The ∼50 Myr Old TOI-942c is Likely on an Aligned, Coplanar Orbit and Losing Mass", *The Astronomical Journal*, 168, 194 [10 citations]

2024: B. J. Hord, E. M.-R. Kempton, T. M. Evans-Soma, D. W. Latham, D. R. Ciardi, et al. **(incl. RAR)**, "Identification of the Top TESS Objects of Interest for Atmospheric Characterization of Transiting Exoplanets with JWST", *The Astronomical Journal*, 167, 233 [42 citations]

2023: S. Blunt, A. Carvalho, T. J. David, C. Beichman, J. K. Zink, et al. **(incl. RAR)**, "Overfitting Affects the Reliability of Radial Velocity Mass Estimates of the V1298 Tau Planets", *The Astronomical Journal*, 166, 62 [57 citations]

2020: A. Albert, R. Alfaro, C. Alvarez, J. C. Arteaga-Velázquez, K. P. Arunbabu, et al. **(incl. RAR)**, "Constraints on the Emission of Gamma-Rays from M31 with HAWC", *The Astrophysical Journal*, 893, 16 [5 citations]

TESS-Keck Survey publications (560 citations)

I contributed substantially to the TKS infrastructure by co-developing the sample construction algorithm (Chontos et al., 2022, AJ), constructing the nightly target lists (298 calendar nights), and co-leading the RV fitting of the HIRES/APF data for all TKS systems (Polanski et al., 2024, ApJS).

2025: J. Van Zandt, et al. **(incl. RAR)**, "The TESS-Keck Survey. XXIV. Outer Giants May Be More Prevalent in the Presence of Inner Small Planets", *The Astronomical Journal*, 169, 235 [18 citations]

2024: D. Pidhorodetska, et al. **(incl. RAR)**, "The TESS-Keck Survey. XXII. A Sub-Neptune Orbiting TOI-1437", *The Astronomical Journal*, 168, 135 [3 citations]

2024: S. Lange, et al. **(incl. RAR)**, "The TESS-Keck Survey. VII. A Superdense Sub-Neptune Orbiting TOI-1824", *The Astronomical Journal*, 167, 282 [4 citations]

2024: A. S. Polanski, et al. **(incl. RAR)**, "The TESS-Keck Survey. XX. 15 New TESS Planets and a Uniform RV Analysis of All Survey Targets", *The Astrophysical Journal Supplement Series*, 272, 32 [72 citations]

2024: A. Desai, et al. **(incl. RAR)**, "The TESS-Keck Survey. XVIII. A Sub-Neptune and Spurious Long-period Signal in the TOI-1751 System", *The Astronomical Journal*, 167, 194 [5 citations]

2024: M. L. Hill, et al. **(incl. RAR)**, "The TESS-Keck Survey. XIX. A Warm Transiting Sub-Saturn-mass Planet and a Nontransiting Saturn-mass Planet Orbiting a Solar Analog", *The Astronomical Journal*, 167, 151 [3 citations]

2024: C. Beard, et al. **(incl. RAR)**, "The TESS-Keck Survey. XVII. Precise Mass Measurements in a Young, High-multiplicity Transiting Planet System Using Radial Velocities and Transit Timing Variations", *The Astronomical Journal*, 167, 70 [17 citations]

2024†: A. Chontos, et al. **(incl. RAR)**, "The TESS-Keck Survey XXI: 13 New Planets and Homogeneous Properties for 21 Subgiant Systems", arXiv:2402.07893 [15 citations]

2023: J. M. Akana Murphy, et al. **(incl. RAR)**, "The TESS-Keck Survey. XVI. Mass Measurements for 12 Planets in Eight Systems", *The Astronomical Journal*, 166, 153 [18 citations]

2023: M. G. MacDougall, et al. **(incl. RAR)**, "The TESS-Keck Survey. XV. Precise Properties of 108 TESS Planets and Their Host Stars", *The Astronomical Journal*, 166, 33 [30 citations]

2023: J. Van Zandt, et al. **(incl. RAR)**, "TESS-Keck Survey. XIV. Two Giant Exoplanets from the Distant Giants Survey", *The Astronomical Journal*, 165, 60 [19 citations]

2022: M. G. MacDougall, et al. **(incl. RAR)**, "The TESS-Keck Survey. XIII. An Eccentric Hot Neptune with a Similar-mass Outer Companion around TOI-1272", *The Astronomical Journal*, 164, 97 [8 citations]

2022: A. Chontos, et al. **(incl. RAR)**, "The TESS-Keck Survey: Science Goals and Target Selection", *The Astronomical Journal*, 163, 297 [51 citations]

2022: E. V. Turtelboom, et al. **(incl. RAR)**, "The TESS-Keck Survey. XI. Mass Measurements for Four Transiting Sub-Neptunes Orbiting K Dwarf TOI-1246", *The Astronomical Journal*, 163, 293 [17 citations]

2022: P. A. Dalba, et al. **(incl. RAR)**, "The TESS-Keck Survey. VIII. Confirmation of a Transiting Giant

Planet on an Eccentric 261 Day Orbit with the Automated Planet Finder Telescope", *The Astronomical Journal*, 163, 61 [38 citations]

2022: J. Lubin, et al. **(incl. RAR)**, "TESS-Keck Survey. IX. Masses of Three Sub-Neptunes Orbiting HD 191939 and the Discovery of a Warm Jovian plus a Distant Substellar Companion", *The Astronomical Journal*, 163, 101 [38 citations]

2021: M. G. MacDougall, et al. **(incl. RAR)**, "The TESS-Keck Survey. VI. Two Eccentric Sub-Neptunes Orbiting HIP-97166", *The Astronomical Journal*, 162, 265 [16 citations]

2021: N. Scarsdale, et al. **(incl. RAR)**, "TESS-Keck Survey. V. Twin Sub-Neptunes Transiting the Nearby G Star HD 63935", *The Astronomical Journal*, 162, 215 [16 citations]

2021: F. Dai, et al. **(incl. RAR)**, "TKS X: Confirmation of TOI-1444b and a Comparative Analysis of the Ultra-short-period Planets with Hot Neptunes", *The Astronomical Journal*, 162, 62 [46 citations]

2021: L. M. Weiss, et al. **(incl. RAR)**, "The TESS-Keck Survey. II. An Ultra-short-period Rocky Planet and Its Siblings Transiting the Galactic Thick-disk Star TOI-561", *The Astronomical Journal*, 161, 56 [50 citations]

2020: F. Dai, et al. **(incl. RAR)**, "The TESS-Keck Survey. III. A Stellar Obliquity Measurement of TOI-1726 c", *The Astronomical Journal*, 160, 193 [33 citations]

2020: P. A. Dalba, et al. **(incl. RAR)**, "The TESS-Keck Survey. I. A Warm Sub-Saturn-mass Planet and a Caution about Stray Light in TESS Cameras", *The Astronomical Journal*, 159, 241 [43 citations]

Publications contributed to as observer (1,203 citations)

I have contributed to a number of works as a result of observing on Keck/HIRES, Keck/KPF, and through my membership on the KPF Instrument Team.

2026: E. Y. Zhang, F. Dai, A. W. Howard, S. P. Halverson, H. Isaacson, et al. **(incl. RAR)**, "Spin-Orbit Alignment of Two Neptune-size Planets Younger Than 500 Myr: TOI-560 b and TOI-5082 b", *The Astronomical Journal*, 172, 201

2026: A. Householder, F. Dai, A. Kesseli, A. W. Howard, S. Halverson, et al. **(incl. RAR)**, "The Keck Planet Finder SURFS-UP Survey. I. Transmission Spectroscopy of WASP-76 b", *The Astronomical Journal*, 172, 48 [5 citations]

2026: M.-T. Wang, F. Dai, H.-G. Liu, K. Masuda, A. W. Howard, et al. **(incl. RAR)**, "TOI-4495: A Pair of Aligned, Near-resonant Sub-Neptunes That Likely Experienced Overstable Migration", *The Astronomical Journal*, 171, 135 [5 citations]

2025: C. L. Brinkman, L. M. Weiss, D. Huber, R. A. Lee, J. Kolecki, et al. **(incl. RAR)**, "The Compositions of Rocky Planets in Close-in Orbits Tend to Be Earth-like", *The Astronomical Journal*, 170, 109 [24 citations]

2025: H.-Y. Teng, F. Dai, A. W. Howard, S. Halverson, H. Isaacson, et al. **(incl. RAR)**, "Stellar Obliquity of the Ultra-short-period Planet System HD 93963", *The Astronomical Journal*, 170, 51 [7 citations]

2025: R. A. Lee, F. Dai, A. W. Howard, S. Halverson, J. Gomez Barrientos, et al. **(incl. RAR)**, "TOI-6324 b: An Earth-mass Ultra-short-period Planet Transiting a Nearby M Dwarf", *The Astrophysical Journal Letters*, 983, L36 [6 citations]

2024: J. Zhang, D. Huber, L. M. Weiss, J. W. Xuan, J. A. Burt, et al. **(incl. RAR)**, "A Testbed for Tidal Migration: The 3D Architecture of an Eccentric Hot Jupiter HD 118203 b Accompanied by a Possibly Aligned Outer Giant Planet", *The Astronomical Journal*, 168, 295 [15 citations]

2024: J. Lubin, E. A. Petigura, J. Van Zandt, C. Beard, F. Dai, et al. **(incl. RAR)**, "The HD 191939 Exoplanet System is Well Aligned and Flat", *The Astronomical Journal*, 168, 196 [10 citations]

2024: H. Isaacson, A. W. Howard, B. Fulton, E. A. Petigura, L. M. Weiss, et al. **(incl. RAR)**, "The California Legacy Survey. V. Chromospheric Activity Cycles in Main-sequence Stars", *The Astrophysical Journal Supplement Series*, 274, 35 [25 citations]

2024: F. Dai, A. W. Howard, S. Halverson, J. Orell-Miquel, E. Pallé, et al. **(incl. RAR)**, "An Earth-sized Planet on the Verge of Tidal Disruption", *The Astronomical Journal*, 168, 101 [22 citations]

2024: N. Saunders, S. K. Grunblatt, A. Chontos, F. Dai, D. Huber, et al. **(incl. RAR)**, "TESS Giants Transiting Giants. VI. Newly Discovered Hot Jupiters Provide Evidence for Efficient Obliquity Damping after

the Main Sequence", *The Astronomical Journal*, 168, 81 [15 citations]

2024†: S. W. Yee, E. A. Petigura, H. Isaacson, A. W. Howard, S. Blunt, et al. **(incl. RAR)**, "Additional Doppler Monitoring Corroborates HAT-P-11c as a Planet", *Research Notes of the American Astronomical Society*, 8, 187 [4 citations]

2024: P. A. Dalba, S. R. Kane, H. Isaacson, B. Fulton, A. W. Howard, et al. **(incl. RAR)**, "Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. IV. Long-term Doppler Spectroscopy for 11 Stars Thought to Host Cool Giant Exoplanets", *The Astrophysical Journal Supplement Series*, 271, 16 [13 citations]

2024: A. Householder, L. M. Weiss, J. E. Owen, H. Isaacson, A. W. Howard, et al. **(incl. RAR)**, "Investigating the Atmospheric Mass Loss of the Kepler-105 Planets Straddling the Radius Gap", *The Astronomical Journal*, 167, 84 [1 citation]

2023: F. Dai, K. C. Schlaufman, H. Reggiani, L. Bouma, A. W. Howard, et al. **(incl. RAR)**, "A Mini-Neptune Orbiting the Metal-poor K Dwarf BD+29 2654", *The Astronomical Journal*, 166, 49 [16 citations]

2023: E. Knudstrup, D. Gandolfi, G. Nowak, C. M. Persson, E. Furlan, et al. **(incl. RAR)**, "Radial velocity confirmation of a hot super-Neptune discovered by TESS with a warm Saturn-mass companion", *Monthly Notices of the Royal Astronomical Society*, 519, 5637 [13 citations]

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